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Product Design:

“The REAL SELL Is In The SOFTWARE And Its TRANSACTIONAL MODEL”

We see a lot of startups and small design teams working on Internet of Things projects that aim to let people control household appliances from a smartphone. Is that really where the potential lies? Ralf Buehler, senior vice president, sales and marketing, element14, takes a look at this and other insights for successful product design in a conversation with Dilin Anand of EBY

Q. What is the biggest opportunity for electronics engineers today?

A. It is one thing being able to turn your lights off from your smartphone, but what really makes this work is a step further, where technology responds to users' normal living patterns, rather than merely providing them with the opportunity to control things around them through a different mechanism. It is a big hardware opportunity but it is probably a much bigger software ecosystem opportunity.

A lot of next-generation companies in this space will not be the ones existing today but will result in new entrepreneurial ideas. This means that engineers or inventors with great ideas have a chance of making it big.

Q. What are some popular areas of engineering R&D today?

A. In the Internet of Things, the trend is about connecting to something and controlling it via a communications setup. While simply sensing what was happening in a remote area was a very big part of things, the segment is developing rapidly. Today, people

are less content with simply sensing and are thinking about developing their system to do the next big thing—autonomously controlling a machine using what it learns from the various sensors connected to it.

It is no longer about simply sending a message, but more about using actuators to act upon the messages it sends and receives. Power is at the heart of challenges that any wireless device faces. Energy harvesting and other methods to improve the uptime of devices is a big area of research.

Q. What is the biggest R&D challenge for those entering this space?

A. The current situation is such that the problem that needs to be solved will probably need a piece of hardware that is not specified as yet. The big challenge then is that, engineers need to learn how to leverage new hardware technology as soon as possible without having to build it from scratch while they are still playing around with a proof of concept.

Hardware should not be a

limitation but a tool or platform that engineers can rapidly build their business case on. Once the business case is proven, engineers can come back to the hardware world to figure out how to get their hardware into mass production.

Q. How has the importance of hardware evolved in the last couple of years?

A. When you look at the hardware trends over the last three years, you will notice that there were a lot of concepts that did not even exist before. Fitness trackers were non-existent until some people realised that these could be made into a business. I would say that it was more of a hardware play than a software one.

On the other hand, a majority of Internet of Things solutions outside of industrial applications are truly more of a software ecosystem development than hardware. This is true even when we look at things like Amazon's Alexa. There is a piece of hardware but it is kind of almost a throwaway product because the real sell is in the software and its transactional model. **EBY**